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PRODUCT REQUIREMENTS DOCUMENT (PRD)

UrjaMitra AI (ఉర్జామిత్ర)

Technical Architecture, Multi-Agent Engine & System Specifications

INITIATIVE	PROBLEM STATEMENT	CORE FRAMEWORK	DATE & STATUS
AIKAM EnergyTech Challenge	**885fa752 (Billing Assistance)**	**LangGraph + FastAPI + Python**	**October 2026 | Tech Baseline**

1. PRODUCT OVERVIEW & ARCHITECTURAL AXIOM

UrjaMitra AI is an enterprise-grade, distributed multi-agent conversational copilot engineered for Telangana power utilities (TGSPDCL and TGNPDCL). It automates personalized electricity bill explanation, decodes telescopic tariff brackets, verifies meter reading photos, and provides seamless dispute resolution in Telugu, Hindi, and English.

> **Architectural Axiom: Zero-Hallucination Deterministic Engine**
> In government utility billing, LLM hallucination on monetary sums is strictly intolerable. UrjaMitra AI enforces a strict separation: **Natural Language Understanding** is performed by LLM agents via LangGraph; whereas **Billing Arithmetic & Tariff Calculation** is executed exclusively by a **100% deterministic Python TGERC Tariff Engine** implementing gazette formulas.

2. FULL-STACK SYSTEM ARCHITECTURE

INGESTION LAYER
WhatsApp Business API
Web & Mobile Responsive UI
1912 Voice Telephony Bridge
DISCOM Officer Portal

AGENTIC BRAIN
LangGraph Supervisor
Agent 1: Slab Deconstructor
Agent 2: OCR Meter Audit
Agent 3: Spike & Solar Advisor
Agent 4: SAP Dispute Ticket

DETERMINISTIC CORE
Python TGERC Tariff Engine
Gruha Jyothi Subsidy Rule
Provisional Bill Adjuster
Hourly Load Analyzer

DATA & INTEGRATIONS
PostgreSQL 16 (ACID DB)
pgvector / Qdrant (RAG)
Redis Cluster (Cache)
TGSPDCL SAP IS-U BAPI

3. DEEP TOPIC INDEXING MATRIX

Topic ID	Component Specification	Scope Category	System Owner	Technical Deliverable
`PRD-01`	Architecture Tiers & Clean Design	Full-Stack Core	Solutions Architect	Microservices Blueprint
`PRD-02`	Multi-Agent Orchestration	Agentic Framework	AI/ML Lead	LangGraph State Engine
`PRD-03`	Deterministic TGERC Tariff Engine	Core Algorithm	Backend Engineer	Python Tariff Math Module
`PRD-04`	Multimodal OCR & Meter Vision	Vision Ingestion	CV Engineer	Digit & Bill Parser
`PRD-05`	Multilingual Vernacular UX	Frontend & Speech	UI/UX & Voice Lead	Telugu/Hindi Web & Voice

Topic ID	Component Specification	Scope Category	System Owner	Technical Deliverable
`PRD-06`	Data Schemas & Persistence	Database & Cache	Data Architect	Postgres + Redis Schemas
`PRD-07`	SAP IS-U BAPI Connectors	Enterprise B2G	Integration Lead	REST/JSON CRM Gateway
`PRD-08`	FinOps & Token Economics	Cost Governance	Cloud FinOps Lead	Multi-tier Model Router

4. DETAILED AGENT SPECIFICATIONS (LANGGRAPH)

Agent 1: Telescopic Slab Deconstructor

Decomposes confusing monthly bills into itemized brackets (0–100, 101–200, 201–300, 301–400, >400 units). Resolves the Gruha Jyothi subsidy calculation: if units ≤ 200, applies zero-bill credit; if units > 200, explains the exact rupee variance factors in plain Telugu and English.

Agent 2: Multimodal Bill OCR & Meter Reading Audit

Ingests mobile camera photos of paper bills and live meter LCD screens. Applies perspective correction, extracts USC Number, Previous KWh, Present KWh, and Status (01 Normal, 02 Door Lock). Automatically calculates variance: `Calculated Units = (Present - Previous) × MF`. If live meter photo shows reading lower than billed reading, flags Average Billing discrepancy.

Agent 3: Proactive Consumption & Spike Advisory

Segments consumption by Time-of-Day (Morning Peak, Solar Hours, Evening Peak, Night Baseload). Detects heatwave cooling spikes and calculates subsidy proximity: `Delta = 200 - Projected Units`. Provides actionable tips in local vernacular and computes rooftop solar feasibility under PM Surya Ghar Muft Bijli Yojana.

Agent 4: Autonomous Dispute Resolution & SAP Ticket Agent

Assembles cryptographic dispute evidence (image hash, OCR confidence, timestamp, difference delta). Automatically generates a **Provisional Revised Bill** according to TGERC guidelines and dispatches a structured JSON payload to the DISCOM SAP IS-U BAPI complaint queue with human-in-the-loop Section Officer sign-off.

5. DETERMINISTIC TGERC TARIFF ENGINE SPECIFICATION

The Python tariff service implements exact Telangana Retail Supply Tariff Order rules:

```
# Official TGERC LT-I Domestic Tariff Schedules
Band A (Total Consumption <= 100 Units/Month):
  - 0 to 50 Units:     ₹1.95 / Unit
  - 51 to 100 Units:   ₹3.10 / Unit

Band B (Total Consumption 101 to 200 Units/Month):
  - 0 to 100 Units:    ₹3.40 / Unit
  - 101 to 200 Units:  ₹4.80 / Unit

Band C (Total Consumption > 200 Units/Month - Telescopic):
  - 0 to 100 Units:    ₹5.10 / Unit       - 301 to 400 Units:  ₹9.50 / Unit
  - 101 to 200 Units:  ₹7.70 / Unit       - 401 to 800 Units:  ₹10.00 / Unit
  - 201 to 300 Units:  ₹9.00 / Unit       - Above 800 Units:   ₹11.00 / Unit

Fixed Charges:     ₹10 - ₹15 per kW contracted load / month
Customer Charges:  ₹40 to ₹80 / month based on consumption tier
Electricity Duty:  ₹0.06 / unit billed
Gruha Jyothi:      100% Subsidy Waiver if tagged eligible & consumption <= 200 Units
```

6. LLM TOKEN ECONOMICS & FINOPS COST PROJECTIONS

Deployment Tier	Daily Active Users	Monthly Inquiries	Model Tier	Est. USD / Month	Est. INR / Month
Pilot (1 Circle)	1,500	45,000	Flash + Haiku	$38.50	**₹3,200**
City Scale (Hyd)	35,000	1,050,000	Tiered Routing	$890.00	**₹74,000**
Full State Scale	250,000	7,500,000	Cached Tiered	$6,350.00	**₹5,27,000**

*Prompt caching reduces system prompt token costs by up to 78%. Deterministic Python microservice executes all mathematics at zero LLM token cost.

7. AIKAM MILESTONE EXECUTION TIMELINE

Date	Milestone Phase	Deliverable Artifacts
October 09, 2026	Phase 1: Architecture Lock	Finalized BRD, PRD, and AIKAM Proposal Blueprint
October 11, 2026	Phase 2: Core Agentic Engine	Deterministic Tariff Engine + LangGraph backend + 500-Consumer synthetic testbed
October 14, 2026	Phase 3: Omnichannel UI Live	Telugu Web/WhatsApp UI + DISCOM Officer Command Center
October 16, 2026	**Immersion Day Review**	In-person demonstration of live bill deconstruction & SAP ticket generation
October 24, 2026	DevFest Hyderabad	Top 15 Finalists Announcement
October 25, 2026	**Innovation Day Finale**	Grand jury demo lock and presentation to Dept of Energy leadership